Yanhui Guo

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EDUCATION BACKGROUND

(Ph.D.) McMaster University **Hamilton, ON, Canada**
Computer Science, Image/Video Restoration, 2D/3D Computer Vision, LLM and Generative AI

(B.Sc./M.Sc.) Huazhong University of Science and Technology **Wuhan, China**
Artificial Intelligence and Automation

My research interests lie in computer vision and machine learning, especially in image restoration, video understanding, and 2D/3D content generation. Since joining the industry, I have been working on search and recommendation systems, while exploring SOTA multimodal generative technologies.

PROFESSIONAL EXPERIENCE

8+ YOE in industy w/ focus on Deep Learning, Computer Vision, Search, and Recommendation

Amazon **Seattle, United States**
Applied Scientist (Tech Lead) *April. 2024 - Present*

- Prime Video - Personalization & Discovery
  - Led cross-domain recommendation systems leveraging customer behaviors across multiple Amazon product categories to build LLM-based retrieval and customer understanding models.
  - Developed high-throughput data pipeline and augmentation strategies for multimodal catalog (text, image, and video) table to boost content understanding models.
  - Built a personalized video thumbnail diffusion model pipeline, integrating recommendation signals and user behaviour feedback, improving engagement with an increase in CTR by xx%.
- Core Search - Central Machine Learning
  - Built ranking models and conducted A/B experiments to enhance the search experience for global Amazon customers. Developed recommendation models to power various Amazon pages.
  - Led a 4-member team to improve relevance rates in Amazon Search using LLMs, reducing irrelevant results by xx bps and boosting operational revenue by xx%.
  - Built LLM-driven multilingual and multimodal substitute recommendation models for Amazon's global stores, increasing product coverage by xx% and operational revenue by xx%.
  - Researched and developed multi-task personalization models (universal user embedding model) in search to support diverse customer use cases, increasing operational revenue by xx%.

Noah's Ark Lab (AI Lab), Canada **Toronto, Canada**
Senior Researcher *Sep. 2023 - April. 2024*

- Led a team to develop an image-to-3D scene product, boosting customer engagement by xx%.
- Research on text-to-3D content generation and 3D reconstruction, e.g., Gaussian splatting.
- Published 3 US patents and papers (e.g., NeurIPS) on content understanding and 3D generation.

Amazon **Seattle, United States**
Applied Scientist *June. 2023 - Sep. 2023*

- Researched prompt tuning for large language models, with results published at NAACL. Developed attribute extraction models to improve valid product's coverage across large-scale catalogs.

Noah's Ark Lab (AI Lab), Canada **Toronto, Canada**
Researcher *Feb. 2022 - June. 2023*

- Conducted research on video understanding and video search, focusing on large-scale representation learning and retrieval efficiency. Led a team to win the runner-up in the CVPR 2022 ActivityNet Challenge.

NetEase Games, AI Lab **Hangzhou, China**
Artificial Intelligence Engineer *July. 2019 - Jan. 2020*

- Developed AI technologies in the gaming industry.

PUBLICATIONS

Natural Language Processing

[1] **Yanhui Guo**, Shaoyuan Xu, Jinmiao Fu, Bryan Wang. “Q-Tuning: Continual Queue-based Prompt Tuning for Language Models”, (**NAACL 2024**)(Paper Link).

Search and Recommendation

[2] Junting Wang, ... **Yanhui Guo**, Multi-modal Relational Item Representation Learning for Substitutable and Complementary Recommendation (**ACM SIGIR Conference 2026**)

[3] **Yanhui Guo**, Dazhou Yu, and et al., ”Satisfying Complex User Needs: M^3 Agent for Conversational Multi-Item Recommendation” (**Amazon Machine Learning Conference 2025**).

[4] Mingdai Yang, **Yanhui Guo**, and et al., “PCL: Prompt-based Continual Learning for User Modeling in Recommender Systems”, (**International World Wide Web Conference 2025**)(Paper Link).

Computer Vision

[5] **Yanhui Guo**, Chenghuan Guo, Yan Gao, Yi Sun, ”Learning by Taking Notes: Memory-Guided Continual Learning for Generative Multimodal Models” (**ICCV 2025 MMFM4**)

[6] Binh M Le, ... **Yanhui Guo** and et al., ”QID: Efficient query-informed ViTs in data-scarce regimes for OCR-free visual document understanding” (**CVPR 2025 MULA**)

[7] **Yanhui Guo**, **Second Prize** in Challenge on Image Super-Resolution ($\times 4$) **(CVPR NTIRE 2025)**.

[8] **Yanhui Guo**, Fangzhou Luo, Xiaolin Wu. “ Learning Degradation Independent Representations for Camera ISP Pipelines”, (**CVPR 2024**)(Paper Link).

[9] Fangzhou Luo, **Yanhui Guo**, and Xiaolin Wu. ”AND: Adversarial Neural Degradation for Learning Blind Image Super-Resolution”, (**NeurIPS 2023**)(Paper Link).

[10] **Yanhui Guo**, Fangzhou Luo, Shaoyuan Xu. “Self-Supervised Face Image Restoration with a One-Shot Reference”, (**ICASSP 2024, Oral**)(Paper Link).

[11] **Yanhui Guo**, Peng Dai, Juwei Lu and Li Cheng. “Refining Implicit Neural Action Field for Temporal Action Localization”, (**Runner-up reward, CVPR 2022 ActivityNet**, US patent,)(Paper Link).

[12] **Yanhui Guo**, Xiao Shu and Xiaolin Wu. “Data Acquisition for Dual-reference Deep Learning of Image Super-Resolution”, (**Transactions on Image Processing (TIP)**)(Paper Link).

[13] Fangzhou Luo, **Yanhui Guo** and Xiaolin Wu. “Functional Neural Networks for Parametric Image Restoration Problems”, (**NeurIPS 2021**)(Paper Link).

[14] **Yanhui Guo**, Xi Zhang and Xiaolin Wu. “Deep Multi-modality Soft-decoding of Very Low Bit-rate Face Videos”, 2020 ACM International Conference on Multimedia (**ACM MM 2020**)(Paper Link).

Generative AI

[15] Hengkang Wang, ... **Yanhui Guo**, et al., ”Temporal-Consistent Video Restoration with Pre-trained Diffusion Models” (**AAAI 2026**)(Paper Link).

[16] Lingjing Kong,...**Yanhui Guo**, et al., ”Learning to Compose the Unseen Combinations: Compositional Generalization through Hierarchical Concept Models” (**CVPR 2026**).

[17] **Yanhui Guo**, Xinxin Zuo, Peng Dai, and et al., ”Decorate3D: Text-Driven High-Quality Texture Generation for Mesh Decoration in the Wild”, (**NeurIPS 2023**, two US patents,)(Project, Paper Link).

Others

- Journal/Conference Reviewer: Journal/Conference Reviewer: Amazon AMLC, ICML, NeurIPS, ECCV, ICME, WACV, CVPR, WWW, IEEE Transactions on Image Processing.